

Peter Chengming Zhang

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Peter Zhang is a pharmacist, technology professional, and PhD student working across healthcare, artificial intelligence, and entrepreneurship. His doctoral research at the University of Toronto's Leslie Dan Faculty of Pharmacy examines how clinicians become practicing entrepreneurs, and what that means for patients, practitioners, and health systems. He also holds a PharmD and an MBA.

EDUCATION

- 2023– Doctor of Philosophy in Pharmaceutical Sciences
Leslie Dan Faculty of Pharmacy, University of Toronto
- 2020–22 Master of Business Administration
Rotman School of Management, University of Toronto
- 2017–21 Doctor of Pharmacy with Honours
Leslie Dan Faculty of Pharmacy, University of Toronto
- 2013–17 Honours Bachelor of Science with High Distinction
Faculty of Arts & Science, University of Toronto

RESEARCH EXPERIENCE

Leslie Dan Faculty of Pharmacy, University of Toronto

- 2023– PhD Student
Supervisors: Dr. Zubin Austin & Dr. Certina Ho
Doctoral research on interventions in clinician entrepreneurship education.
- 2019–20 Project Researcher
Supervisor: Dr. Carolyn Cummins
Investigated roles of JMJD6 and RBM39 in alternative splicing pathways mediated by the glucocorticoid receptor and the ARGLU1 protein.
- 2018–19 Lab Assistant
Supervisor: Dr. Carolyn Cummins
Conducted laboratory techniques including PCR, RNA extraction, and gel electrophoresis.

The Reach Alliance, University of Toronto

- 2026– Co-Mentor
- 2018–20 Researcher
Supervisor: Dr. Anita McGahan
Qualitative field study in New Delhi and Uttar Pradesh, India; examined strategies for overcoming vaccine hesitancy in hard-to-reach populations in the context of polio elimination.

University of York

- 2017 Project Researcher
Supervisor: Dr. Paul Kaye
Cell staining, flow cytometry, and RNA extraction to investigate monocyte activation in the context of leishmaniasis.

Temerty Faculty of Medicine, University of Toronto

2016–17

Undergraduate Researcher

Supervisor: Dr. Dana Philpott

qPCR, genotyping, ELISA, bone marrow harvesting, and RNA extraction to investigate dendritic cell differentiation in Crohn's disease.

Ontario Institute for Studies in Education, University of Toronto

2014–15

Undergraduate Researcher

Supervisor: Dr. Becky Chen

Psychometric testing of elementary school students; investigated language acquisition differences between bilingual and multilingual learners.

PEER-REVIEWED PUBLICATIONS

JOURNAL ARTICLES & BOOK CHAPTERS

1. Zhang, P.C., & Tadrous, M. (2025). Empowering pharmacists in Canada's drug shortage response. *Canadian Pharmacists Journal / Revue des Pharmaciens du Canada*. doi:10.1177/17151635251365666
2. Ng, K.K.Y., Matsuba, I., & Zhang, P.C. (2024). RAG in health care: A novel framework for improving communication and decision-making by addressing LLM limitations. *NEJM AI*, 2(1). doi:10.1056/Alra2400380
3. Zhang, P.C., Matinnia, C., & Austin, Z. (2024). To maximise impact, hospital pharmacists need to increase visibility. *European Journal of Hospital Pharmacy*, 31(3), 185–186. doi:10.1136/ejhpharm-2024-004137
4. Ng, K.K.Y., Zhang, P.C. (2023). Advancing medical affairs capabilities and insight generation through machine learning techniques. *Journal of Pharmaceutical Policy and Practice*, 165(16). doi:10.1186/s40545-023-00670-w
5. Zhang, P.C., Austin, Z. (2023). To succeed as health care innovators, pharmacists need to cultivate an entrepreneurial mindset. *Canadian Pharmacists Journal*, 156(3), 100–111. doi:10.1177/17151635231164622
6. Ho, C., Zhang, P.C. (2022). Patient Safety from a Pharmacy Perspective. In *Encyclopedia of Evidence in Pharmaceutical Public Health and Health Services Research in Pharmacy*. Springer, Cham. doi: 10.1007/978-3-030-50247-8_72-1
7. Daud, A., Hussein, I.M., Zhang, P.C., Ahmed, Y., Trac, J., Vujovic, N., Rizvi, S.F., & Kuzyk, P.R. (2021). 3D-Printed personal protective face shields during the COVID-19 pandemic: A survey of Canadian frontline workers. *Cureus*. doi: 10.7759/cureus.18141
8. Zhang, P.C. (2021). The future of pharmacy is intertwined with digital health innovation. *Canadian Pharmacists Journal*. doi:10.1177/17151635211044474
9. Bellatin, A., Hyder, A., Rao, S., Zhang, P.C., & McGahan, A.M. (2021). Overcoming vaccine deployment challenges among the hardest to reach: Lessons from polio elimination in India. *BMJ Global Health*, 6(4), e005125. doi: 10.1136/bmjgh-2021-005125
10. Zhang, P.C., Ahmed, Y., Hussein, I.M., et al. (2020). Optimization of community-led 3D printing for the production of protective face shields. *3D Printing in Medicine*, 6, 35. doi:10.1186/s41205-020-00089-y

TEACHING

LECTURER — UNIVERSITY OF TORONTO

2023–

Change Management (PHM310)

Leslie Dan Faculty of Pharmacy · Undergraduate

2024–25

AI in the Pharmaceutical Industry (BTC2030)

Leslie Dan Faculty of Pharmacy · Graduate/Undergraduate

2025	Power & Leadership (PHM215) Leslie Dan Faculty of Pharmacy · Undergraduate
2024	Entrepreneurship (PHM327) Leslie Dan Faculty of Pharmacy · Undergraduate
2022–	Introduction to Financial Literacy; Business Planning (PHM215) Leslie Dan Faculty of Pharmacy · Undergraduate
2021–	Microbiology in the Clinical Setting (HMB312) Faculty of Medicine & Pharmacy · Undergraduate
2018–23	Antimicrobial Stewardship (NEW116) New College · Undergraduate
2017–23	Vaccine Hesitancy (NEW106) New College · Undergraduate

INSTRUCTIONAL ROLES

2021–22	Clinical Instructor Leslie Dan Faculty of Pharmacy, University of Toronto
2020	Scholarship of Teaching and Learning Research Assistant Leslie Dan Faculty of Pharmacy, University of Toronto
2018–21	Pharmacy Lab Teaching Assistant Leslie Dan Faculty of Pharmacy, University of Toronto
2021	Scholar, Managing Consumer Value Rotman School of Management, University of Toronto
2016–17	Senior Peer Mentor First Year Learning Communities, University of Toronto
2014–16	Peer Mentor First Year Learning Communities, University of Toronto

KNOWLEDGE TRANSLATION & PUBLIC SCHOLARSHIP

INVITED SPEAKING

2026	“Advancing Safe and Responsible AI Adoption in Community Pharmacy Practice” Network for Improving Health Systems, Leslie Dan Faculty of Pharmacy
2026	“AI in the Modern Workforce” Bausch Health
2025	“Artificial Intelligence, e-Therapies and Virtual Care” Canadian Psychiatric Association
2024	“The AI Revolution in Healthcare” Pharmacy Changemakers, University of Toronto
2024	“COVID-19 Vaccine Update” British Columbia Pharmacy Association
2024	“3D Printing Face Shields” CAPSI Professional Development Week
2024	“Designing & Evaluating Impact” The AFS Youth Assembly

- 2023 "Harnessing the Power of Communication and Storytelling for Business Growth"
Healthcare Businesswomen's Association

INTERVIEWS & PODCASTS

- 2024 "Collaborative Care: A Pharmacist's Insights on Research and Medicine"
Canadian Federation of Medical Students
- 2024 "Artificial intelligence holds promise for pharmacy research and care"
Leslie Dan Faculty of Pharmacy, University of Toronto
- 2021 "Why choose a PharmD-MBA at the Leslie Dan Faculty of Pharmacy?"
Leslie Dan Faculty of Pharmacy, University of Toronto

KNOWLEDGE TRANSLATION (73 OP-EDS & ARTICLES)

1. Gan, J.Q., Tadrous, M., & Zhang, P. "Cross-border prescriptions: What Canadian pharmacists need to know." *Canadian Healthcare Network*. April 13, 2026.
2. Zhang, P. "Three ways AI enables the entrepreneurship process." *Canadian Healthcare Network*. April 2, 2026.
3. Sachdeva, N. & Zhang, P. "Why (and How) Teachers Should Use Generative AI in the Classroom." *Faculty Focus*. March 4, 2026.
4. Zhang, P. & Danchuk-Lauzon, B. "Bridging intergenerational tensions in the workplace." *CareerWise | CERIC*. February 13, 2026.
5. Zhang, P. "It's never too early to join a board!" *Canadian Healthcare Network*. January 26, 2026.
6. Zhang, P. "Universities face a financial crisis. The key to survival is the scholar-entrepreneur." *The Globe and Mail*. January 21, 2026.
7. Zhang, P. "MaaTRx: When a pharmacist entrepreneur builds a novel drug shortage response platform." *Canadian Healthcare Network*. January 2, 2026.
8. Samani, M., Elsayes, S. & Zhang, P. "Pharmacists: Bridging the gap between AI and patient-centred care." *Healthy Debate*. November 28, 2025.
9. Zhang, P. "What pharmacists can learn from Asepha's \$4M seed fund raising." *Canadian Healthcare Network*. September 12, 2025.
10. Zhang, P. & Tadrous, M. "As pharmacists, we must be empowered to act in the face of Canada's painkiller shortage." *Toronto Star*. August 10, 2025.
11. Zhang, P. "The future of healthcare depends on investments in clinician entrepreneurship education." *Canadian Healthcare Network*. July 31, 2025.
12. Zhang, P. "How to leverage clinical skills in non-traditional careers." *Canadian Healthcare Network*. June 24, 2025.
13. Zhang, P. "Breaking into the pharmaceutical industry: a guide for pharmacy students." *Canadian Healthcare Network*. April 22, 2025.
14. Zhang, P. "Everyone wins when we invest in pharmacist entrepreneurs." *Canadian Healthcare Network*. April 2, 2025.
15. Daud, A. & Zhang, P. "In the face of growing uncertainty, Canada needs to double down on innovation." *Healthy Debate*. April 1, 2025.
16. Zhang, P. "Will an MBA help clinicians grow?" *Canadian Healthcare Network*. March 17, 2025.
17. Pitkis, A., Tami, P. & Zhang, P. "The importance of health-system navigators." *Healthy Debate*. January 28, 2025.
18. Tian, J.L.H., Shafaqat, A., Yaseen, S. & Zhang, P. "5 Ways to Propel Inclusive Climate Action." *Mastercard*. September 17, 2024.

19. Seladurai, M. & Zhang, P. "Cybersecurity attacks highlight the need for patients to keep their medication records accessible." *Healthy Debate*. September 15, 2024.
20. Tian, J.L.H., Shafaqat, A., Yaseen, S. & Zhang, P. "Climate Resilience at the Margins: Five Key Lessons from Last Mile Populations." *Global Policy Journal*. July 29, 2024.
21. Sivanthan, L., Pitkis, A. & Zhang, P. "British Columbia found the key to attracting more family doctors, Ontario should do the same." *Toronto Star*. May 5, 2024.
22. Ng, K.K.Y. & Zhang, P. "AI Disruption is Coming. Are Healthcare Professionals Ready?" *MedCity News*. April 1, 2024.
23. Peters, C. & Zhang, P. "Governments should pay the high price of gene therapy." *Healthy Debate*. February 26, 2024.
24. Zhang, P. "Canadian pharmacists have an important role this RSV season." *Canadian Healthcare Network*. December 19, 2023.
25. Battey, J. & Zhang, P. "Prescribing community for mental health – lessons from Australia." *Healthy Debate*. December 3, 2023.
26. Shafaqat, A., Tan, D. & Zhang, P. "Canada must invest in global leadership programs within higher education institutions." *The Hill Times*. November 6, 2023.
27. Verhage, S., Tian, J. & Zhang, P. "Untapped potential: Young leaders have the tools to help address health-care challenges." *Healthy Debate*. October 23, 2023.
28. Younis, M. & Zhang, P. "Pharmacist prescribers fill important gaps in the Canadian healthcare system." *Healthy Debate*. September 24, 2023.
29. Bala, S., Goobie, G. & Zhang, P. "Clearing the air: fighting for a breathable future." *Toronto Star*. September 1, 2023.
30. Ahmed, A., Yang, D. & Zhang, P. "Naloxone kits in TTC stations a necessary tool to address overdoses." *Toronto Star*. August 5, 2023.
31. Philbert, N. & Zhang, P. "Free menstrual hygiene products are on the way, if you work at a bank, an airline or the public service." *Toronto Star*. June 4, 2023.
32. Mathew, L. & Zhang, P. "Preventative health care: Why privately funded programs are reluctant to fund it." *Toronto Star*. May 6, 2023.
33. Zhang, P. "Synthetic data may revolutionize clinical trials – and save lives." *Healthy Debate*. April 3, 2023.
34. Zhang, P. & Daud, A. "Modernized home care can help stabilize health costs." *The Globe and Mail*. March 22, 2023.
35. Pitkis, A. & Zhang, P. "The benefits of giving include tax credits." *Toronto Star*. March 20, 2023.
36. Pandey, A. & Zhang, P. "Overcoming language barriers key to providing health-care." *Toronto Star*. March 5, 2023.
37. Zhang, P. "Ask a pharmacist before you split tablets for your children." *Toronto Star*. November 21, 2022.
38. Yang, J., Ahmed, A. & Zhang, P. "Internet-based mental health care deserves a boost." *Ottawa Citizen*. November 21, 2022.
39. Yang, J., Ahmed, A. & Zhang, P. "Vaccine hesitancy is reviving long dead diseases." *Healthy Debate & National Post*. September 12, 2022.
40. Zhang, P. & Tadrous, M. "How pharmacist prescribing can alleviate health-care system pressures." *Policy Options*. May 31, 2022.
41. Zhang, P. "The Rotman Review: How Trade Missions Teach MBAs To Think International." *Poets & Quants*. May 6, 2022.
42. Daud, A. & Zhang, P. "Elon Musk, social media giants, must prioritize mental health and well-being." *Toronto Star*. May 2, 2022.
43. Zhang, P. "The Rotman Review: Navigating the Future of Digital Marketing." *Poets & Quants*. March 31, 2022.

44. Zhang, P. "The Rotman Review: A Year In The Life Of A MBA Board Fellow." *Poets & Quants*. February 23, 2022.
45. Zhang, P. "Bleeding disorders in women need more attention." *The Hamilton Spectator*. February 8, 2022.
46. Zhang, P. "The Rotman Review: How An MBA Shapes Success In Finance." *Poets & Quants*. February 4, 2022.
47. Zhang, P. "Despite a place on pharmacy shelves, homeopathy has no role in patient care." *Toronto Star*. December 31, 2021.
48. Zhang, P. "The Rotman Review: Why Sustainable Finance Is Critical To MBAs." *Poets & Quants*. December 5, 2021.
49. Ahmed, A. & Zhang, P. "The dangers of antimicrobial drug resistance must not be ignored." *Healthy Debate & National Post*. November 15, 2021.
50. Zhang, P. "The Rotman Review: Building Your Own Time 'Portfolio'." *Poets & Quants*. November 3, 2021.
51. Zhang, P. "The Rotman Review: How Academic Research Makes MBAs More Valuable." *Poets & Quants*. September 22, 2021.
52. Ahmed, A. & Zhang, P. "As election day looms, voters have the chance to be heard on the federal funding of health care." *Toronto Star*. September 10, 2021.
53. Zhang, P. "The Rotman Review: Preparing For The Future Of Machine Learning." *Poets & Quants*. August 22, 2021.
54. Garland, M., Ahmed, A. & Zhang, P. "Communication must become an essential component of the science education curriculum." *Healthy Debate*. August 18, 2021.
55. Zhang, P. & McGahan, A.M. "To overcome vaccine hesitancy, governments need to show citizens what's in it for them." *Rotman Insights Hub*. August 16, 2021.
56. Ahmed, A. & Zhang, P. "To move beyond the pandemic, Canada needs to look globally." *The Hill Times*. July 29, 2021.
57. Ahmed, Y., Kwan, J. & Zhang, P. "Clinicians as community advocates: interplays of vaccines and society." *University of Toronto Medical Journal*. June 24, 2021.
58. Yao, A., Zhang, P. & Ho, C. "When COVID-19 meets allergy season." *Hospital News*. May 27, 2021.
59. Zhang, P. & Daud, A. "Rigorous scrutiny of COVID-19 vaccines should bolster, not erode, confidence in immunization safety." *Toronto Star*. May 3, 2021.
60. Tang, J. & Zhang, P. "Should more of our foods be fortified with vitamin D?" *The Hamilton Spectator*. April 19, 2021.
61. Ahmed, Y., Daud, A. & Zhang, P. "COVID-19 and insomnia: Pills are not the answer." *Healthy Debate*. April 16, 2021.
62. Zhang, P. "Prioritize one-shot COVID-19 vaccines for the hard to reach." *Toronto Star*. April 7, 2021.
63. Zhang, P. & Hu, M. "Mobilizing pharmacists is key to immunization success." *The Hamilton Spectator*. March 5, 2021.
64. Zhang, P. & Ahmed, Y. "Cyberchondria: An unexpected side effect of digitizing medical information." *Toronto Star*. February 16, 2021.
65. Bala, S. & Zhang, P. "To win against COVID-19, we must put vaccine hesitancy behind us." *Ottawa Citizen*. January 4, 2021.
66. Daud, A. & Zhang, P. "Although alarming, mutated strains of COVID-19 are not a cause for panic." *Toronto Star*. January 3, 2021.
67. Zhang, P., Chan, C. & Ho, C. "Antimicrobial Stewardship in the face of COVID-19." *Hospital News*. October 28, 2020.
68. Zhang, P. "Naloxone needs to be in hands of as many people as possible." *The Windsor Star*. October 4, 2020.
69. Zhang, P. "We need naloxone kits in more hands in this opioid crisis." *The Hamilton Spectator*. October 3, 2020.
70. Zhang, P. & Daud, A. "As pandemic forces change, it's a good time for Canadians to quit smoking." *Toronto Star*. September 9, 2020.
71. Zhang, P. "It's time to normalize suicide prevention training at the workplace." *Toronto Star*. August 28, 2020.

72. Zhang, P., Ahmed, Y., Daud, A., Hussein, I., Junghan, J., Khalid, M., Khalid, M., & Shah, S. "Becoming a Pit Crew for Frontline Healthcare Workers during the COVID-19 Pandemic." *CMAJ Blogs*. August 14, 2020.
73. Zhang, P., Teo, V. & Khosravani, H. "When COVID-19 meets flu season." *Toronto Star*. July 14, 2020.

AWARDS & RECOGNITION

HONOURS & SCHOLARSHIPS

2024	Emerging Leader in Pharmacy Practice Award — Leslie Dan Faculty of Pharmacy
2021	Rotman Scholar Award — Rotman School of Management
2021	Student Engagement Award — University of Toronto
2021	Joan E. Foley Quality of Student Experience Award — University of Toronto
2020	Guy Genest Passion for Pharmacy Award — CAPSI
2020	Future of Pharmacy Excellence Award — CAPSI
2020	James D. Fleck Entrance Award — Rotman School of Management
2020	Frank Kwiecien Memorial Scholarship — Leslie Dan Faculty of Pharmacy
2020	COVID-19 Student Engagement Award — University of Toronto
2020	CSHP External Affairs Student Writing Award — Undergraduate Pharmacy Society
2017	Gordon Cressy Student Leadership Award — University of Toronto
2017	Queen Elizabeth II Diamond Jubilee Scholarship — Trinity College, University of Toronto
2017	Terry Buckland Memorial Award — University of Toronto
2017	Health Plus Pharmacy Award — Mackenzie Health Volunteer Association
2017	New College Student Council Leadership Award — New College, University of Toronto
2016	New College Council In-Course Scholarship — New College, University of Toronto
2016	Ruth Reiffenstein Memorial Scholarship — New College, University of Toronto
2016	New College 50th Anniversary Award — New College, University of Toronto

RECOGNITION

2023	Youth Assembly Delegate — AFS Intercultural Programs
2022	100 Best & Brightest MBAs: Class of 2022 — Poets & Quants
2020	3DPPE GTHA Printing Milestone of 20,000 Face Shields — Ontario Minister of Health

BOARD EXPERIENCE

2025–	Board of Directors, Centre for Effective Practice
2022–25	Board of Directors, Hemophilia Ontario
2021–22	Board Fellow, Hemophilia Ontario
2015–16	Board of Directors, University of Toronto Students' Union

PROFESSIONAL EXPERIENCE

ODALA

- 2025– Senior Business Acceleration Partner & Value Delivery Leader
- 2023–25 Business Acceleration Partner

Twig Fertility

- 2025–26 Clinical Pharmacist

Southlake Health

- 2021– Clinical Pharmacist
- 2018 System Technician Intern

Johnson & Johnson Innovative Medicine

- 2023 Marketing Manager, Cardiovascular & Metabolism
- 2023 Marketing Associate, Cardiovascular & Metabolism
- 2022–23 Marketing Associate, Neuroscience
- 2021 Marketing Associate Intern, Dermatology

3DPPE GTHA

- 2020 Co-Founder

ACADEMIC SERVICE & COMMUNITY

Leslie Dan Faculty of Pharmacy

- 2025– Graduate Department Academic Appeals Committee
- 2025– Alumni Awards Selection Committee
- 2018–19 Curriculum and Assessment Committee

Canadian Society of Hospital Pharmacists

- 2019–20 Membership Committee
- 2018–19 Communications Committee

New College Student Council, University of Toronto

- 2017–18 Speaker
- 2016–17 Graduating Year Representative
- 2015–16 Social Director

New College Orientation, University of Toronto

- 2016 General Planning Executive
- 2015 Director of Training & Scheduling

2014 Orientation Leader

Mackenzie Health Richmond Hill Hospital

2010–17 Information Desk Volunteer Supervisor

2015 Emergency Department Volunteer

Undergraduate Pharmacy Society

2018–19 Webmaster

Poets & Quants

2021–22 Columnist

City of Markham

2012–17 Program Instructor